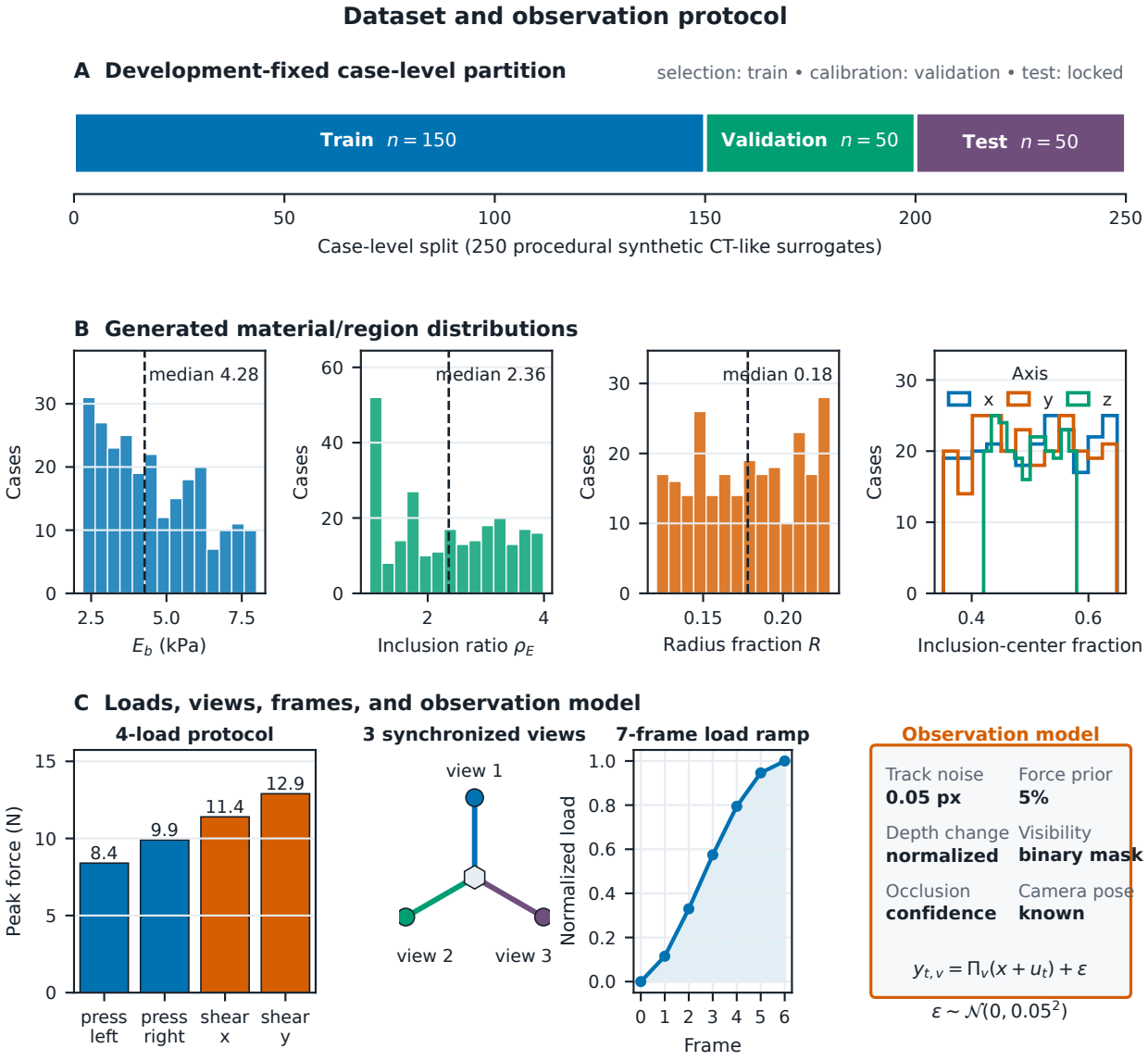
Dataset and observation protocol
A Development-fixed case-level partition
selection: train • calibration: validation • test: locked
Train n = 150
Validation n = 50
Test n = 50
0
50
100
150
200
250
Case-level split (250 procedural synthetic CT-like surrogates)
B Generated material/region distributions
median 4.28
Cases
E_b (kPa)
median 2.36
Inclusion ratio ρ_E
median 0.18
Radius fraction R
Axis
x
y
z
Inclusion-center fraction
C Loads, views, frames, and observation model
4-load protocol
Peak force (N)
8.4
9.9
11.4
12.9
press left
press right
shear x
shear y
3 synchronized views
view 1
view 2
view 3
7-frame load ramp
Normalized load
Frame
Observation model
Track noise
0.05 px
Force prior
5%
Depth change
normalized
Visibility
binary mask
Occlusion
confidence
Camera pose
known
$y_{t,v} = \Pi_v(x + u_t) + \varepsilon$
$\varepsilon \sim \mathcal{N}(0, 0.05^2)$